

light is meant for roadway application, then it needs to meet roadway illumination specification, i.e., the optical pattern has to meet the uniformity of illumination and glare rating. To meet the roadway regulation specification, how to let the light output project on the large roadway area is also important.

To meet the specification of mounting height of 6 metres, correct distribution of the light output along the 18m roadway length was necessary. So, as there existed a specific length, the roadway needs to install streetlight. Then you need more streetlights to apply to meet the roadway specification and avoid the zebra pattern appearing.

This case study for an AOP streetlight installed on 8m pole height, and projected on 32m roadway length, claims that optical pattern can largely reduce the streetlight installation quantities as on a specific roadway length.

Power mechanism and industry design. In AOP's streetlight, they also considered the moving and installation convenience. In the power space, they have installed a socket, so the operator can easily install the city cable into the socket and this process is the same as the traditional lamp installation. By the way, the bow-shaped design in the front of the luminaire can let the operator easily move. This luminaire is dust- and water-resistant and its rating reaches the design goals for LED light module and driver power design."

Continuing development

The efficiency and light output of LEDs are doubling about every 36 months since the 1960s—in a way similar to Moore's law. The advances are generally attributed to the parallel development of other semiconductor technologies and advances in optics and material science. This trend is normally called Haitz's Law after Dr Roland Haitz.

The first high-brightness blue LED was demonstrated by Shuji Nakamura of Nichia Corporation and was based on InGaN.

Around 1999, Philips Lumileds introduced power LEDs capable of

continuous use at one watt. These LEDs used much larger semiconductor die sizes to handle the large power inputs.

The efficiency of LED-based lighting is measured by its light output per unit power input. White LEDs quickly matched and overtook the efficiency of standard incandescent lighting systems. In 2002, Lumileds made 5W LEDs available with a luminous efficiency of 18–22 lumens per watt.

In September 2003, a new type of blue LED was demonstrated by Cree, Inc. to provide 24 mW at 20 milliamperes. This produced a commercially packaged white light giving 65 lm/W at 20 mA, becoming the brightest white LED commercially available at the time, and more than four times as efficient as standard incandescent bulbs. In 2006, Cree demonstrated the prototype of a white LED with a record luminous efficiency of 131 lm/W at 20 mA.

Seoul Semiconductor had planned for 135 lm/W by 2007 and 145 lm/W by 2008. Nichia Corporation has developed a white-light LED with luminous efficiency of 150 lm/W at a forward current of 20 mA.

It should be noted that high-power (≥ 1 W) LEDs are necessary for general lighting applications. Typical operating currents for these devices begin at 350 mA.

Cree is reported to have developed an LED prototype achieving 161 lumens/watt at room temperature. The total output was 173 lumens, and the correlated colour temperature was reported to be 4689K.

Market forecasts

Over the past few years, the light-output efficiency of LEDs has largely risen from 35 lm/W (2006) to 100 lm/W. With their efficiency increasing, LED products that replace traditional lighting have been well anticipated, especially in high-power-consumption products.

Global demand for electric lighting is forecast to reach \$40 billion in 2012 on annual gains of more than seven per cent. The emerging economies of China, India and Russia are projected

to be the fastest growing markets for electric lighting products through 2012. Driven by rising incomes and rapidly growing non-residential construction, these three nations will account for 40 per cent of the additional demand generated between 2007 and 2012. The greatest gains will come from China.

LED lighting devices are forecast to become the fastest growing products in every regional market over the forecast period, as these products continue to penetrate both the construction and manufactured goods markets. Gains will be particularly fast in the Asia-Pacific region, where demand for LED lighting is expected to nearly double between 2007 and 2012.

Indian scenario

In India, a number of companies are manufacturing/marketing LED lights in Mumbai, Delhi, Chennai, Kolkata, Bengaluru, Ahmedabad and other major cities of India. These include Binary Opto Electronics Pvt Ltd in Kolkata and Kwaliti Photonics Pvt Ltd in Hyderabad.

GlacialTech, a company engaged in LED lighting business, sees big growth potential. According to a company spokesperson, "The percentage shift from traditional lighting to LED lighting depends on improvements in the LED chip technology and lowering of the manufacturing cost. But, we have already seen this transition happen, and it will be quicker from this year."

GlacialTech expects over 80 per cent CAGR (compound annual growth rate) in the next four years.

White LEDs for home lighting offer great benefits for the far-flung tribal villages of India. This technology was first implemented in the state of Andhra Pradesh to provide low-cost, long-lasting, renewable energy home lighting systems for poor villagers. Small solar panels can be used to produce enough energy to keep these LED lights burning for hours in rural huts. This will trim down the dependence on kerosene lanterns for lighting. ●

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